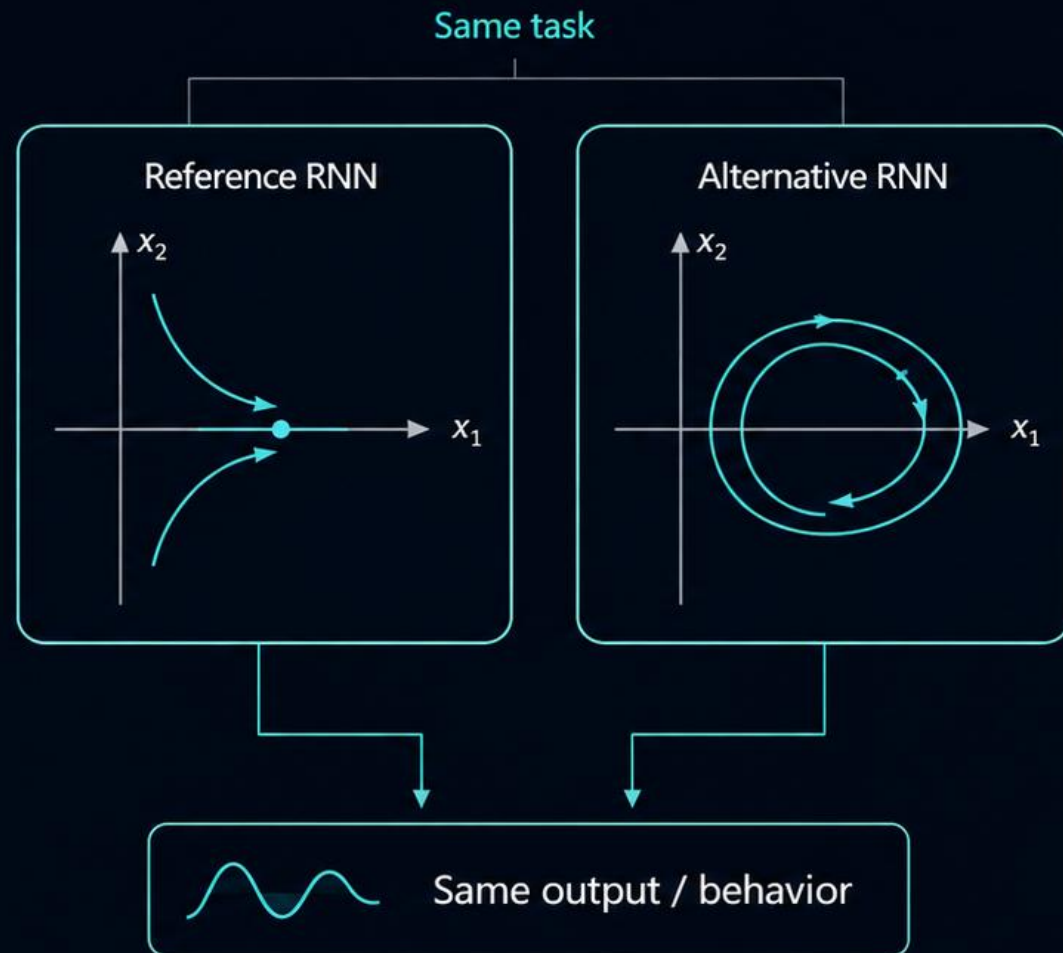


# Discovering Alternative Solutions Beyond the Simplicity Bias in Recurrent Neural Networks

Can the same task be solved by fundamentally different internal dynamics?

William Qian & Cengiz Pehlevan | ICLR 2026



Different internal dynamics can support the same task.

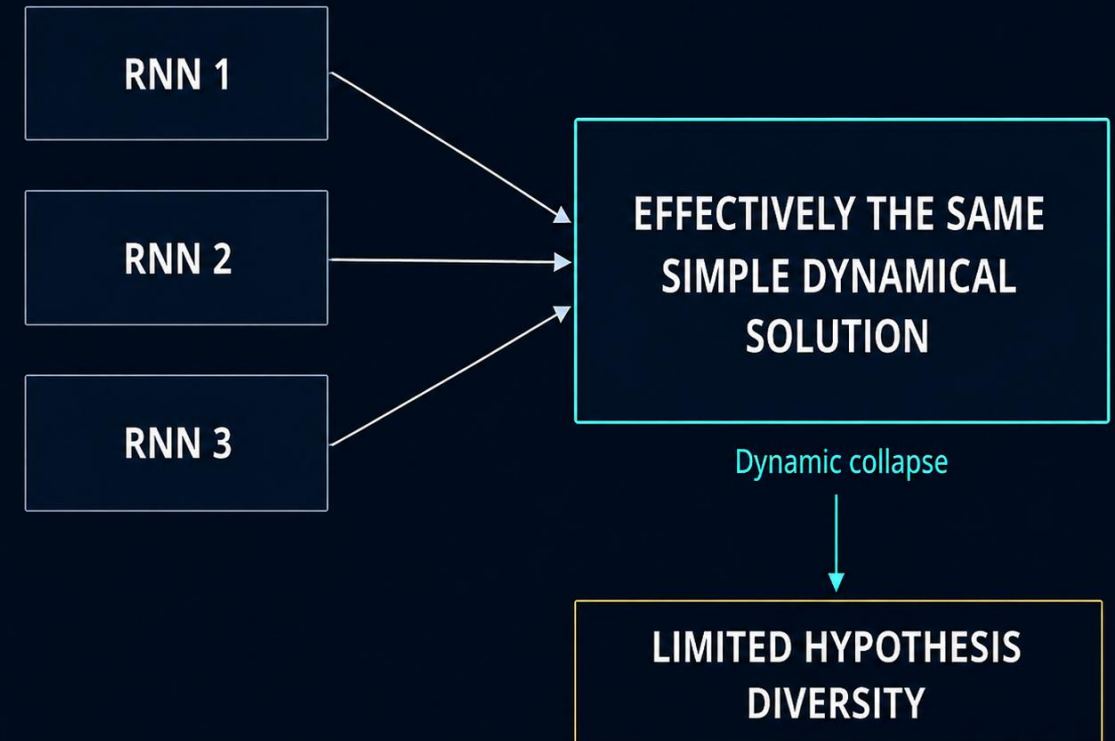
## Why simplicity bias is a neuroscience problem?

Task-trained RNNs are useful if they yield multiple plausible mechanistic hypotheses.

- Standard training favors simple, low-dimensional dynamics.
- Different seeds or gains can converge to the same dynamical motif.
- This creates dynamic collapse: many models, one mechanism.

Core failure mode:  
parameter diversity does not imply mechanism diversity.

SAME TASK · STANDARD TRAINING



## What counts as a genuinely different solution?

A valid alternative must preserve task performance while changing the underlying computation.

### 01 Preserve function

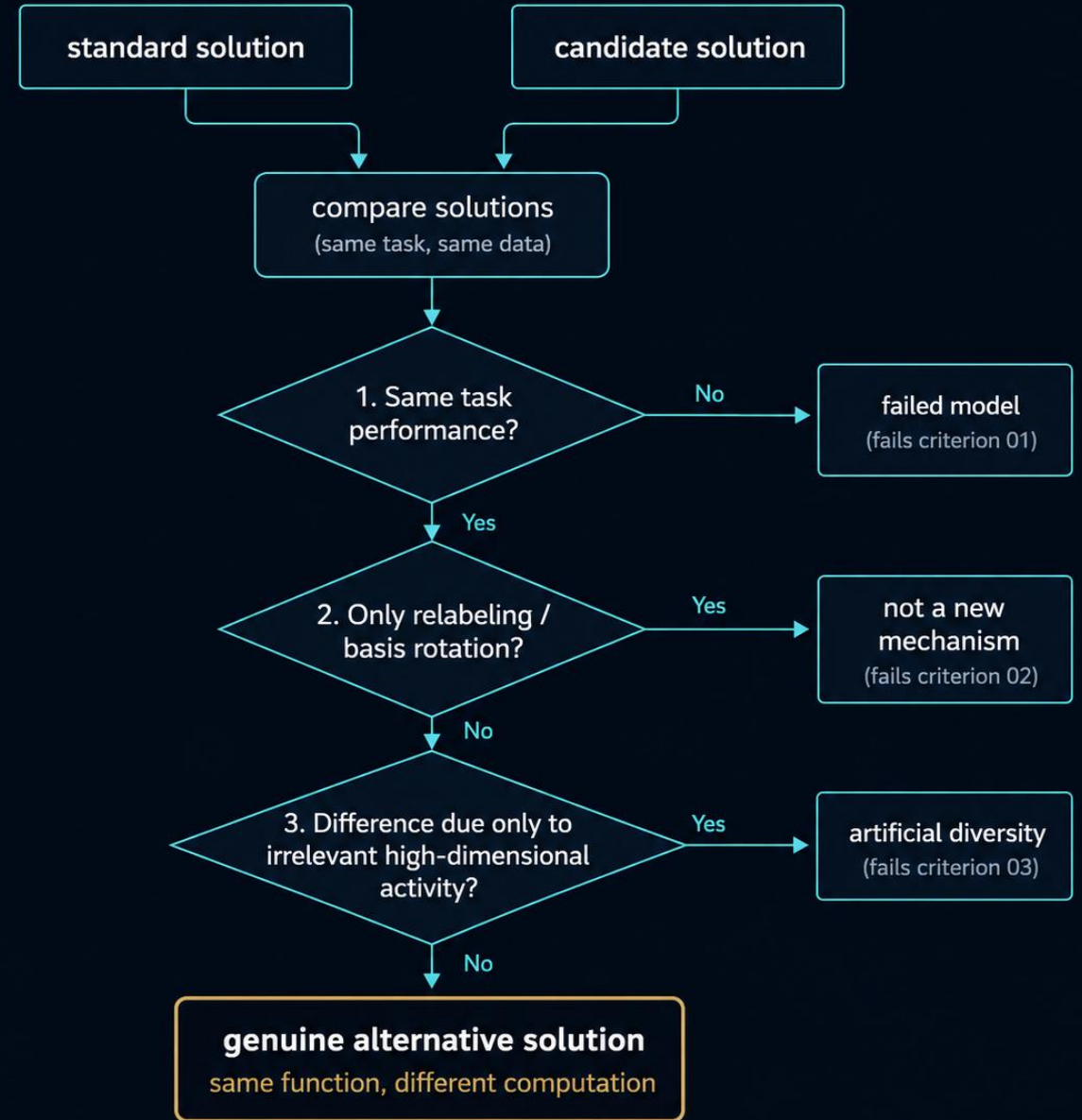
Solve the same task equally well.

### 02 Reject superficial differences

Relabeling neurons or rotating the basis is not a new mechanism.

### 03 Reject trivial dissimilarity

Irrelevant high-dimensional activity must not fake dissimilarity.



Mechanistic diversity is the target.

## Reference RNN and ordinary task training

### Training a standard RNN

Continuous-time rate dynamics

$$\frac{dx}{dt} = -x + W_r r + J_{in} u(t)$$

$$r = \tanh(x), \quad y(t) = J_{out} r(t)$$

Training objective

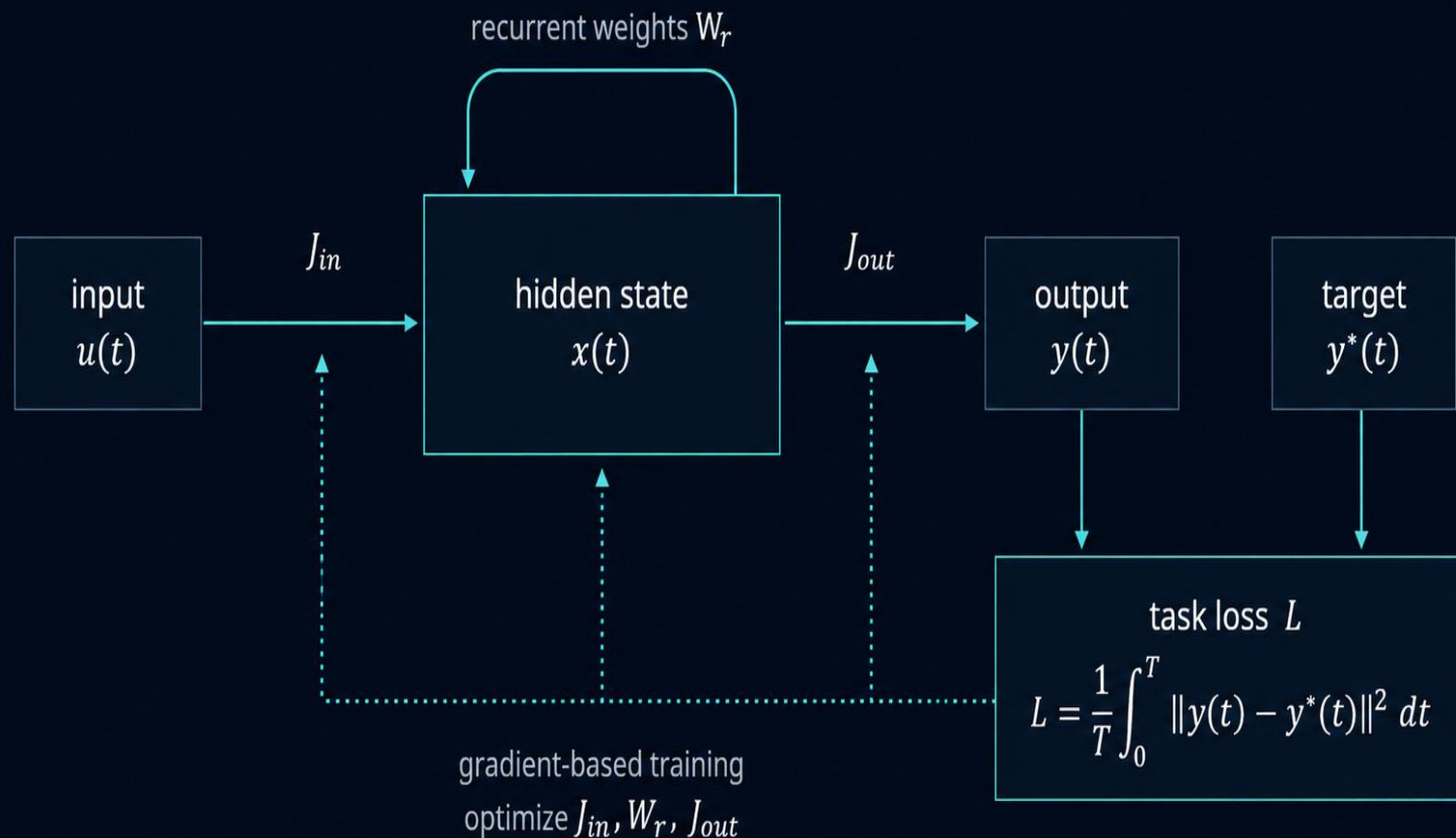
$$L = \frac{1}{T} \int_0^T \|y(t) - y^*(t)\|^2 dt$$

#### Standard comparison population

10 seeds for each initialization gain  
 $g \in \{0.01, 0.5, 1.0, 1.5\}$

Performance is optimized, but solution diversity is not.

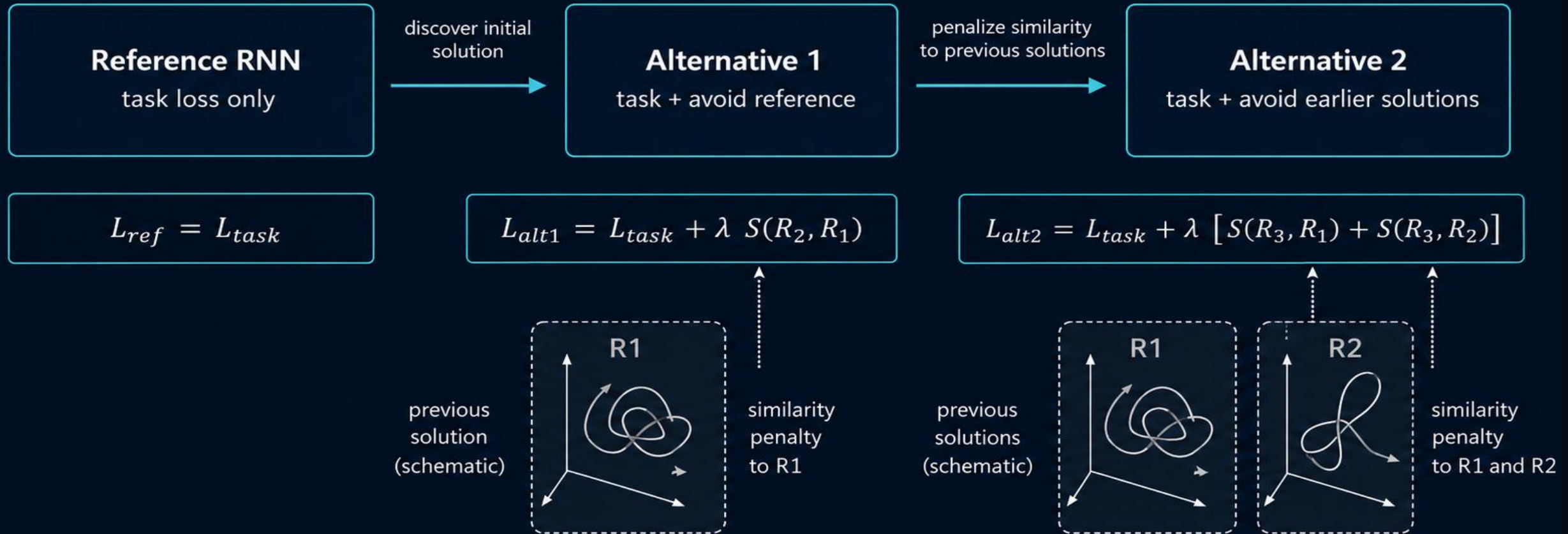
### REFERENCE TRAINING





# INSD: discover a new RNN solution by avoiding previous ones

Task performance is preserved; similarity to earlier neural dynamics is penalized

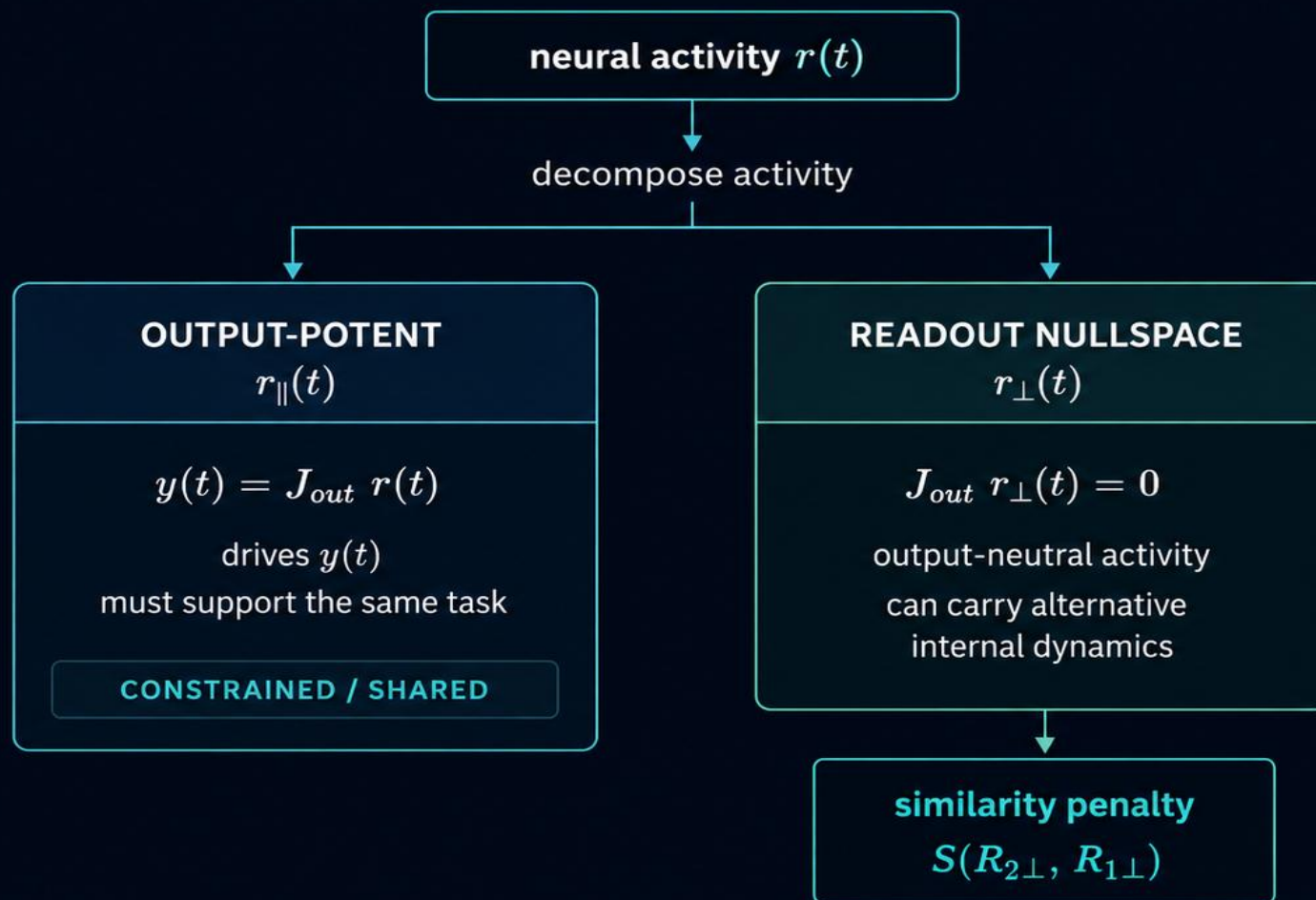


$S(\cdot, \cdot)$ : similarity between neural activity patterns

Same task • progressively different internal dynamics

# Why penalize the readout nullspace?

The output-constrained component must stay similar; INSD therefore penalizes only output-neutral activity.



## Penalty placement

**Use:**  $S(R_{2\perp}, R_{1\perp})$

**Not:**  $S(R_2, R_1)$

Penalize only  
output-neutral activity.

————— **Preserve the shared output computation; diversify the hidden dynamics.** —————

# Linear predictivity is asymmetric by design

- ✓ Can the previous solution be linearly recovered from the new RNN?

$$r^2(X, Y) = 1 - \min_M \frac{\|XM - Y\|^2}{\|Y\|^2}$$

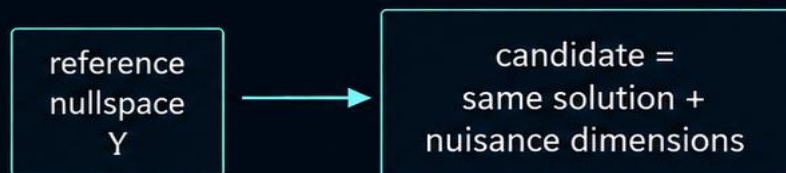
X: new RNN nullspace activity ( $R_{2\perp}$ )

Y: previous/reference nullspace activity ( $R_{1\perp}$ )

$r^2$ : linear predictivity score

High score = previous nullspace activity is still linearly recoverable from the new network.

## Loophole with symmetric distance (schematic)

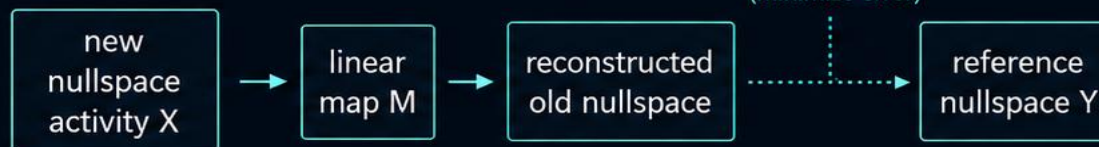


Extra irrelevant activity can make a model look different without changing the underlying mechanism.

✗ Symmetric distance can be fooled.

## Chosen direction

candidate nullspace → reference nullspace



If the previous solution is hidden inside the new nullspace activity, it remains reconstructable and is penalized.

✓ Detects hidden reuse of the old mechanism.

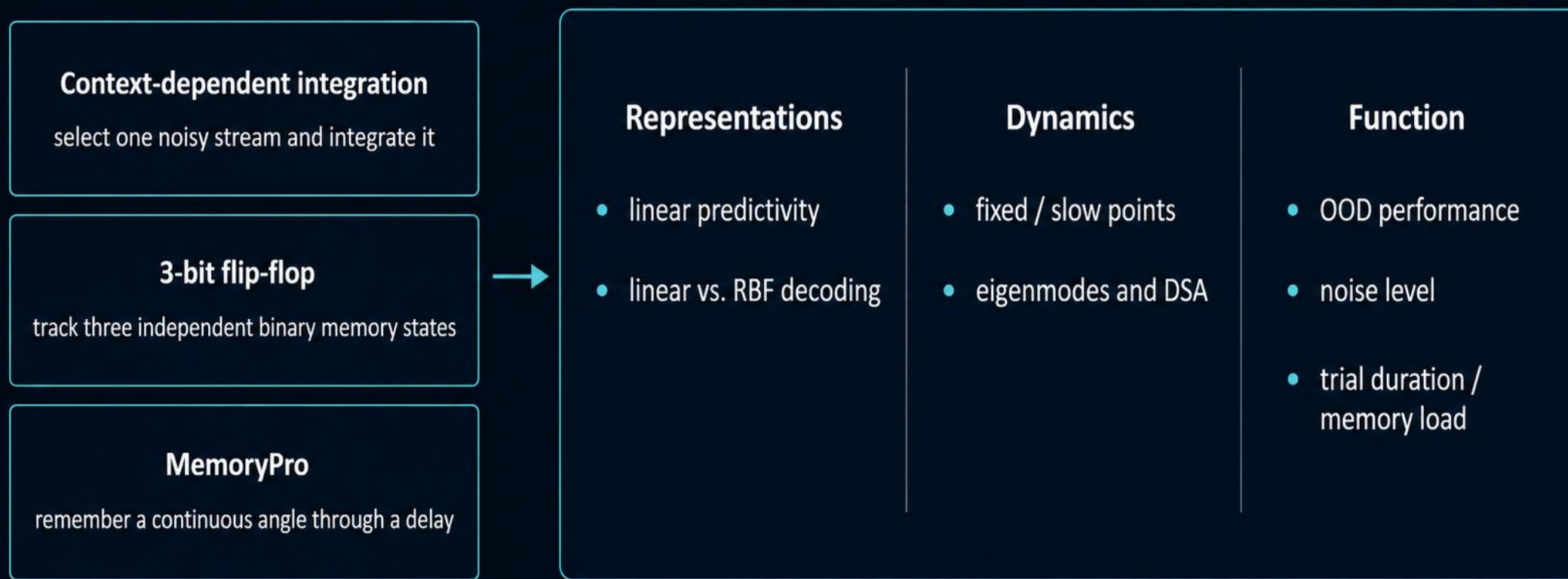
**Penalize recoverability of the old nullspace solution from the new RNN.**

Adding nuisance dimensions cannot fake a new mechanism.



## How the authors rule out cosmetic differences

The paper validates INSD alternatives at the representation, dynamical, and functional levels.

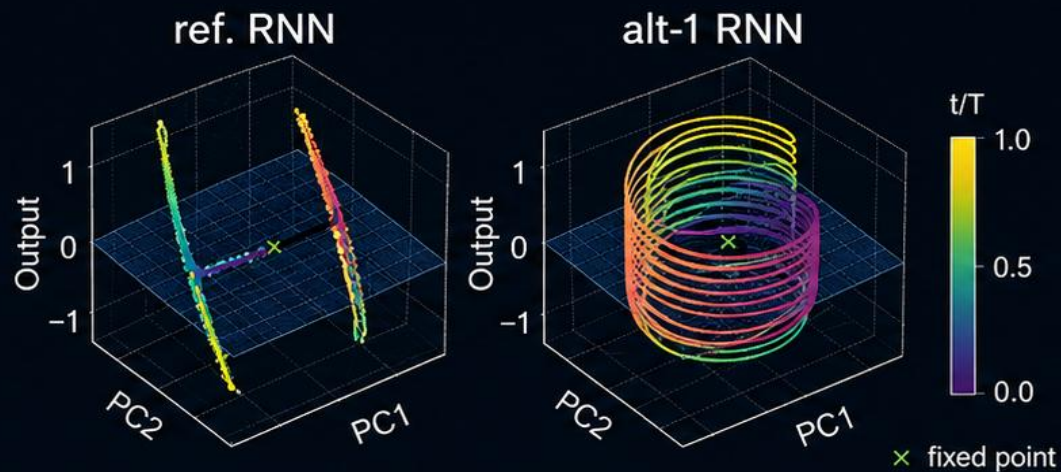


**Validation question: are alternatives merely different representations—or genuinely different mechanisms?**



# INSD changes the dynamical mechanism, not just the plot

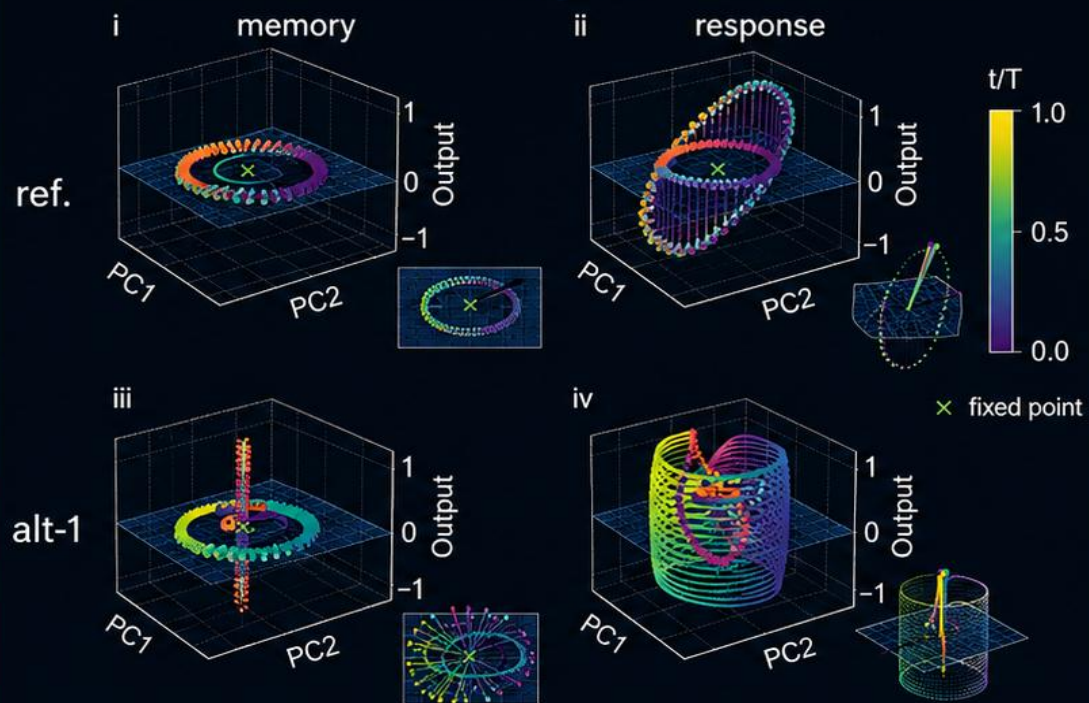
## Context-dependent integration



slow-point line → oscillatory dynamics

- INSD separates from the standard solution.
- Same task variable, different internal code.

## MemoryPro

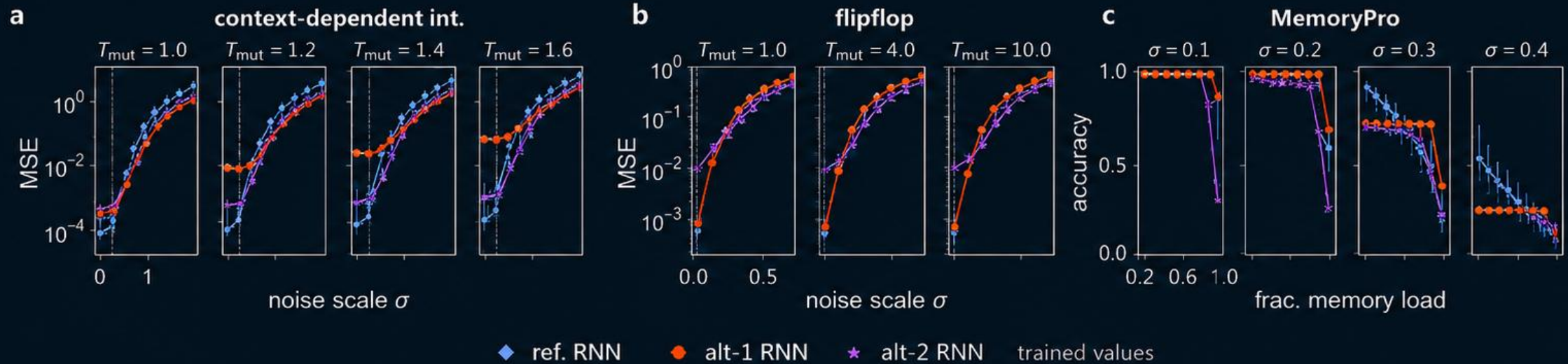


ring attractor → rotating dynamic code

- Persistent memory is preserved.
- Representation changes from static to dynamic.

Same function, different dynamical mechanism.

# Different mechanisms have different robustness profiles



- Standard RNNs usually perform best near the training regime.
- Some INSD solutions outperform under high noise or high memory load.
- **Therefore** INSD is not universally better – it produces functionally distinct trade-offs.

## ⚠ Caveat

Different does not automatically mean more biologically realistic.



## Conclusion

INSD expands the accessible hypothesis space of recurrent neural computation beyond the solutions found by standard training.

# Thank you for you attention

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